P0 Walk-Forward Validation

Point-in-time backtest of price-derived factor blocks | 2025-02-01 to 2025-10-31 | monthly rebalance, 9 dates | generated 2026-08-20T17:35:42

What this report does and does not establish

This is not a validation of Model 5.0. It exercises only the price-derived blocks. Quality, growth and value together carry 70% of the production score and require point-in-time fundamentals that are not yet ingested, so none of them are tested here.

It answers one question from p0.md §7E: **how much of the model's behaviour is momentum?** A momentum-only ranking that performs close to the full model would mean the fundamental blocks are decoration. Read every number below as evidence about momentum and risk, not about Model 5.0.

Two further gaps: these ablations rank **market-wide** where production ranks within sector, and momentum-only omits RS_Sector_6M_Pct (20% of the production momentum block) because the bhavcopy archive carries no sector classification.

Integrity controls

Control	Status
Universe source	Archived NSE bhavcopy per session (point-in-time, includes delisted names)
Survivorship	3384 securities; 522 delisted retained in-sample
Identifier	ISIN-bridged Security_ID; 419 face-value changes linked
Execution	Signal on close of t, fill at open of next confirmed session
Corporate actions	659 split/bonus adjusted, 11580 dividends, 620 unadjustable events excluded
Delisting policy	haircut@0.50
Costs	effective-dated Indian charges + 10bp half-spread + sqrt impact at Rs 1.00 lakh/position
Eligibility	min turnover Rs 20.00 lakh/day, trading frequency >= 0.8, history >= 200 sessions, INE-class only (ETFs excluded)
Parameters	Frozen; no re-optimisation across the window

Eligible universe per rebalance

Metric	Value
Rebalance dates	9
Mean eligible	1602
Min / max eligible	1588 / 1619
Mean considered before filters	2182

Rank information coefficient — Gross of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	9	0.0285	0.0387	78%	-0.0897	0.1204
growth only	3M	9	0.0436	0.0629	78%	-0.0790	0.1092
growth only	6M	9	0.0721	0.0823	89%	-0.0113	0.1409
model 5	1M	9	0.0176	0.0423	67%	-0.2377	0.1906
model 5	3M	9	0.0530	0.1123	67%	-0.2180	0.1940
model 5	6M	9	0.1368	0.2041	89%	-0.1005	0.2444
model 5 gated	1M	9	0.0127	0.0408	67%	-0.2405	0.1917
model 5 gated	3M	9	0.0491	0.1072	67%	-0.2251	0.2064
model 5 gated	6M	9	0.1371	0.1964	89%	-0.1073	0.2433
momentum only	1M	9	-0.0113	0.0127	56%	-0.2457	0.1751
momentum only	3M	9	0.0187	0.0688	67%	-0.2044	0.1155
momentum only	6M	9	0.0800	0.1239	78%	-0.1256	0.1809
quality only	1M	9	0.0229	0.0171	56%	-0.1189	0.1418
quality only	3M	9	0.0379	0.0237	67%	-0.1243	0.1573
quality only	6M	9	0.0775	0.0997	89%	-0.0619	0.1794
random ranking	1M	9	0.0024	0.0002	56%	-0.0329	0.0294
random ranking	3M	9	0.0050	0.0042	56%	-0.0220	0.0352
random ranking	6M	9	0.0072	0.0050	67%	-0.0308	0.0539
simple quality momentum	1M	9	-0.0089	0.0381	56%	-0.2958	0.1932
simple quality momentum	3M	9	0.0209	0.0692	67%	-0.2638	0.1620
simple quality momentum	6M	9	0.1012	0.1541	78%	-0.1483	0.2120
value only	1M	9	0.0290	0.0228	78%	-0.0126	0.0993
value only	3M	9	0.0379	0.0469	89%	-0.0141	0.0567
value only	6M	9	0.0585	0.0567	100%	0.0211	0.0773

Portfolio and bucket evidence — Gross of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	2.00%	2.00%	+0.00%	0.030	-0.78%
equal weight universe	3M	3.76%	3.76%	+0.00%	0.358	-1.74%

equal weight universe	6M	-1.37%	-1.37%	+0.00%	0.224	-3.17%
growth only	1M	2.00%	3.79%	+1.79%	0.479	+0.12%
growth only	3M	3.76%	7.81%	+4.05%	0.515	+0.63%
growth only	6M	-1.37%	4.42%	+5.79%	0.576	+2.17%
model 5	1M	2.00%	1.79%	-0.21%	0.188	+1.04%
model 5	3M	3.76%	5.36%	+1.60%	0.564	+3.16%
model 5	6M	-1.37%	2.89%	+4.26%	0.952	+8.71%
model 5 gated	1M	2.00%	1.71%	-0.30%	0.188	+0.08%
model 5 gated	3M	3.76%	4.18%	+0.42%	0.576	+1.33%
model 5 gated	6M	-1.37%	-0.53%	+0.84%	0.867	+6.80%
momentum only	1M	2.00%	1.07%	-0.93%	0.067	+0.02%
momentum only	3M	3.76%	2.25%	-1.51%	0.661	+1.67%
momentum only	6M	-1.37%	8.59%	+9.96%	0.964	+6.60%
quality only	1M	2.00%	0.81%	-1.20%	0.091	-0.04%
quality only	3M	3.76%	1.10%	-2.66%	0.127	+0.23%
quality only	6M	-1.37%	-2.64%	-1.26%	0.467	+2.48%
random ranking	1M	2.00%	3.23%	+1.23%	0.321	+0.53%
random ranking	3M	3.76%	5.86%	+2.10%	0.224	+1.10%
random ranking	6M	-1.37%	0.85%	+2.22%	0.285	+1.90%
simple quality momentum	1M	2.00%	2.07%	+0.07%	-0.261	-0.02%
simple quality momentum	3M	3.76%	1.55%	-2.21%	0.345	+0.57%
simple quality momentum	6M	-1.37%	-4.17%	-2.80%	0.964	+6.06%
value only	1M	2.00%	3.36%	+1.35%	0.442	+0.22%
value only	3M	3.76%	8.39%	+4.63%	0.648	+0.83%
value only	6M	-1.37%	9.08%	+10.45%	0.758	+5.08%

Rank information coefficient — Net of costs

Spearman correlation between the score on the rebalance date and the forward return. Reported with its distribution because a mean carried by two extraordinary periods is a different claim from one earned consistently. **%+** is the share of periods with a positive IC.

Strategy	Horizon	Periods	Mean	Median	%+	Worst	Best
equal weight universe	1M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	3M	0	n/a	n/a	n/a	n/a	n/a
equal weight universe	6M	0	n/a	n/a	n/a	n/a	n/a
growth only	1M	9	0.0461	0.0603	78%	-0.0755	0.1350
growth only	3M	9	0.0540	0.0751	89%	-0.0711	0.1207
growth only	6M	9	0.0794	0.0903	89%	-0.0044	0.1478
model 5	1M	9	0.0451	0.0632	78%	-0.2171	0.2019
model 5	3M	9	0.0691	0.1312	67%	-0.2061	0.2059

model 5	6M	9	0.1480	0.2186	89%	-0.0872	0.2568
model 5 gated	1M	9	0.0415	0.0648	78%	-0.2195	0.2032
model 5 gated	3M	9	0.0657	0.1267	67%	-0.2130	0.2194
model 5 gated	6M	9	0.1486	0.2114	89%	-0.0937	0.2563
momentum only	1M	9	0.0134	0.0544	56%	-0.2351	0.1819
momentum only	3M	9	0.0334	0.0886	67%	-0.1977	0.1267
momentum only	6M	9	0.0899	0.1274	89%	-0.1185	0.1925
quality only	1M	9	0.0521	0.0488	67%	-0.0900	0.1632
quality only	3M	9	0.0551	0.0423	89%	-0.1097	0.1711
quality only	6M	9	0.0899	0.1129	89%	-0.0463	0.1882
random ranking	1M	9	0.0020	0.0076	56%	-0.0342	0.0268
random ranking	3M	9	0.0049	0.0032	56%	-0.0234	0.0361
random ranking	6M	9	0.0069	0.0041	67%	-0.0316	0.0525
simple quality momentum	1M	9	0.0264	0.0651	56%	-0.2663	0.2099
simple quality momentum	3M	9	0.0418	0.0942	67%	-0.2479	0.1778
simple quality momentum	6M	9	0.1157	0.1698	89%	-0.1316	0.2282
value only	1M	9	0.0123	0.0078	67%	-0.0260	0.0778
value only	3M	9	0.0279	0.0352	89%	-0.0225	0.0471
value only	6M	9	0.0514	0.0491	100%	0.0153	0.0701

Portfolio and bucket evidence — Net of costs

vs universe is the top-20 mean period return minus the eligible universe mean, so it already controls for how the market did. **Monotonicity** is +1 when decile means fall cleanly from best score to worst, 0 when the ranking carries no ordering. **Spread** is top decile minus bottom decile.

Strategy	Horizon	Universe	Top 20	vs universe	Monotonicity	D1-D10
equal weight universe	1M	0.28%	0.28%	+0.00%	0.248	-0.64%
equal weight universe	3M	2.04%	2.04%	+0.00%	0.358	-1.60%
equal weight universe	6M	-3.09%	-3.09%	+0.00%	0.224	-3.03%
growth only	1M	0.28%	2.18%	+1.90%	0.782	+0.65%
growth only	3M	2.04%	6.23%	+4.19%	0.576	+1.17%
growth only	6M	-3.09%	2.88%	+5.97%	0.576	+2.72%
model 5	1M	0.28%	0.56%	+0.28%	0.612	+2.04%
model 5	3M	2.04%	4.11%	+2.07%	0.709	+4.17%
model 5	6M	-3.09%	1.65%	+4.74%	0.952	+9.72%
model 5 gated	1M	0.28%	0.69%	+0.41%	0.746	+1.17%
model 5 gated	3M	2.04%	3.16%	+1.12%	0.648	+2.42%
model 5 gated	6M	-3.09%	-1.54%	+1.55%	0.891	+7.89%
momentum only	1M	0.28%	-0.29%	-0.57%	0.551	+0.73%
momentum only	3M	2.04%	0.89%	-1.16%	0.758	+2.37%
momentum only	6M	-3.09%	7.23%	+10.32%	0.988	+7.31%

quality only	1M	0.28%	-0.35%	-0.63%	0.733	+0.97%
quality only	3M	2.04%	-0.08%	-2.12%	0.430	+1.23%
quality only	6M	-3.09%	-3.87%	-0.78%	0.685	+3.49%
random ranking	1M	0.28%	1.47%	+1.18%	0.176	+0.52%
random ranking	3M	2.04%	4.07%	+2.03%	0.212	+1.09%
random ranking	6M	-3.09%	-0.94%	+2.15%	0.212	+1.89%
simple quality momentum	1M	0.28%	1.11%	+0.83%	0.697	+1.08%
simple quality momentum	3M	2.04%	0.59%	-1.45%	0.758	+1.68%
simple quality momentum	6M	-3.09%	-5.13%	-2.04%	0.964	+7.17%
value only	1M	0.28%	1.35%	+1.06%	0.370	+0.26%
value only	3M	2.04%	6.38%	+4.34%	0.661	+0.85%
value only	6M	-3.09%	7.07%	+10.16%	0.746	+5.10%

Turnover

One-way turnover of a top-20 portfolio per rebalance. 1.0 means the portfolio was fully replaced. High turnover is what converts a real gross edge into no net edge.

Strategy	Mean one-way turnover	Rebalances
equal weight universe	0.106	9
growth only	0.078	9
model 5	0.356	9
model 5 gated	0.494	9
momentum only	0.506	9
quality only	0.100	9
random ranking	0.939	9
simple quality momentum	0.356	9
value only	0.139	9

Fill coverage

Share of scored positions that produced a usable forward return. A dropped position is recorded with a reason rather than silently omitted; a thin horizon here weakens every statistic above it.

Horizon	Coverage	Positions	Main reasons dropped
1M	99.6%	122094	unadjustable_action 441, no_entry_price 36
3M	98.8%	122094	unadjustable_action 1449, no_entry_price 36
6M	97.9%	122094	unadjustable_action 2583, no_entry_price 36

How to read this

The random_ranking row is the calibration. It is a seeded random score, so its IC is what "no signal" measures on this universe and horizon. Any strategy whose IC is not clearly separated from it has not demonstrated anything.

equal_weight_universe scores every name identically, so its rank IC is undefined rather than zero. It is present as the return benchmark from p0.md §7B, not as a ranking.

Judge consistency before magnitude. A mean IC of 0.03 with 70% of periods positive is stronger evidence than a mean of 0.06 driven by two periods. Check the %+ and worst columns before the mean.

Net is the number that matters. If a gross edge disappears net of costs, the ranking may still be real but is not tradable at this position size and turnover.

Known limitations of this run

Restatement bias remains. No fundamentals are used here, so it does not bite yet — but it will once the XBRL panel lands, for any period falling back to current values rather than as-filed ones.

Delisting recovery is an assumption, not an observation. The bhavcopy does not record why a security stopped trading. Re-run with --delisting-strategy zero and last_close to bound how much the headline depends on it.

Period count is modest. Monthly rebalancing over roughly four years is enough to judge cross-sectional rank IC and its stability. It is not enough to claim regime robustness across a full cycle.

Overlapping horizons are not independent. A 6M horizon sampled monthly reuses most of its window, so any t-statistic on per-period ICs overstates significance and is reported as a rough guide only.

Cost assumptions are documented estimates. Brokerage defaults to zero (discount-broker delivery); half-spread and impact coefficient are modelled, not measured. Gross and net are shown separately for that reason.

Next step

Phase B: ingest NSE XBRL filings into a point-in-time fundamental panel keyed on filing timestamp, which unlocks the quality, growth and value blocks and makes a genuine Model 5.0 walk-forward possible. Until then the fundamental 70% of the score remains untested.